

2019 will be a Great and Busy Year for Open Source Artificial Intelligence

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Open source has been eating the software world across multiple industries and has already become the de facto method of developing software, with countless industries - from entertainment to automotive to energy - are increasingly relying on it. Today, more leading-edge software development occurs inside open source communities than ever before, and it is becoming increasingly difficult for proprietary projects to keep up with the rapid pace of development that open source offers.

The AI Attraction to Open Source

Open source is a market accelerant and the advantages it offered and continues to offer across various industry verticals do apply to artificial intelligence (AI), notably:

- It introduces the aspect of standardization around an open technology (standardization around code and not specifications) and that in turn enables better interoperability as solutions are standardized on open technologies
- It offers a neutral environment that fosters collaboration and cross-pollination
- It acts as an innovation multiplier and helps solves problems that can only be solved by fostering collaboration among many players
- It allows sharing the cost of development
- It harmonizes stacks, aligns vendors and qualifies reference architectures
- It enables new business opportunities and lowers the barriers to enter a new domain, a huge value for smaller companies and startups that don't have access to capital nor resources as their larger competitors
- It leads to better products

Open Source is a Market Accelerant

- Standardization and interoperability around open technologies
- Provides a neutral environment that fosters collaboration
- Solves problems that we can only solve together by joining efforts
- Democratizes technology by lowering barriers to enter a new domain
- Drives value creation via its innovation multiplier and network effect
- Allows shared cost of development
- Enables new business opportunities
- Qualifies reference architectures
- Leads to better products

AI is no different than other technology sectors, and is becoming dominated by open source software that brings together multiple collaborators and organizations. Everyone will benefit from the increased innovation that comes from jointly collaborating on machine learning and deep learning technologies and projects.

What does 2019 have for Open Source AI?

More Open Sourcing Activities

More companies will be open sourcing their internal AI, machine learning (ML) and deep learning (DL) projects in order to build communities around those projects and benefit from the open source advantages. It is very hard for a week to pass by without a press release announcing internal AI/ML/DL code being available under an open source license.

Organizations are realizing that the cost of building a platform is high and justifies the open source approach. The real value is in the models, training data and applications.

Collaboration at the platform level will enable companies and communities to harmonize across a set of critical projects that put together will create a comprehensive open source stack in the AI, ML and DL space. Companies who were the first to open source their AI efforts are already seeing early mover advantages, and we expect this trend to accelerate in 2019.

More Projects Hosted Under Foundations

Companies have also realized that opening their code under an open source license is not enough. Structured and neutral governance is required so other organizations are comfortable and encouraged to join and contribute to the project's and become an active member of its community.

One way to achieve that is to host the project under a foundation like The Linux Foundation, which creates a neutral environment for collaboration and acts as a guardian that prevents a single entity from controlling the project. With more than 1,200 companies backing tens of thousands of active developers, The Linux Foundation's projects harness the power of open source development to fuel innovation at unmatched speed and scale.

Why Foundations are a great host to open source projects?

- They offer a legal framework for IP management in which commercial companies can work with open source projects and contributors.
- They offer technical services, such as software repositories and issue tracking, mailing lists, file servers, IRC server, code signing certifications, technical mentorship, etc.
- The support with business operations and governance, which includes financial and banking services, membership management, communications and PR around projects, and event services.

Hosted projects would benefit from the experience of the Foundation in multiple ways:

- **Neutral facilitator and a trust enabler:** Trust that decision makers are unbiased, that the project will stick, that it is worth investing R&D budget, and that the intentions of other participants are true. Neutrality and trust allows projects to thrive. These two factors are required for participants to make durable strategic commitments, to enable participants to form development alliances, and to encourage participants to contribute valuable features that make the project better.
- **–Knowledge, expertise and credibility:** Well-launched project will benefit from startup momentum for a long time. However, this requires help from people who have done it successfully before, knowledge of what works and what doesn't, and existing credibility in launching and managing large scale multi-tenants open source projects. In addition, projects would benefit from the Foundation's services tailored to the project such as IT and infrastructure, marketing and communications, legal support, events, etc.

Data Licensing, Governance, Policies, Privacy and Security

For open source AI to really thrive, there is a need to standardize licensing of data. The Linux Foundation pioneered this effort with the creation of the Community Data License Agreement to enable sharing of data similar to what

open source licenses did to the sharing of source code.

In 2019, we expect further efforts to create data curation projects and open source data sets that can be used to train models. In addition, there will be increased discussion on critical questions related to data security and privacy. Tools and policies will need to be developed to safeguard privacy, anonymize data and remove bias.

Consolidation Bound to Happen

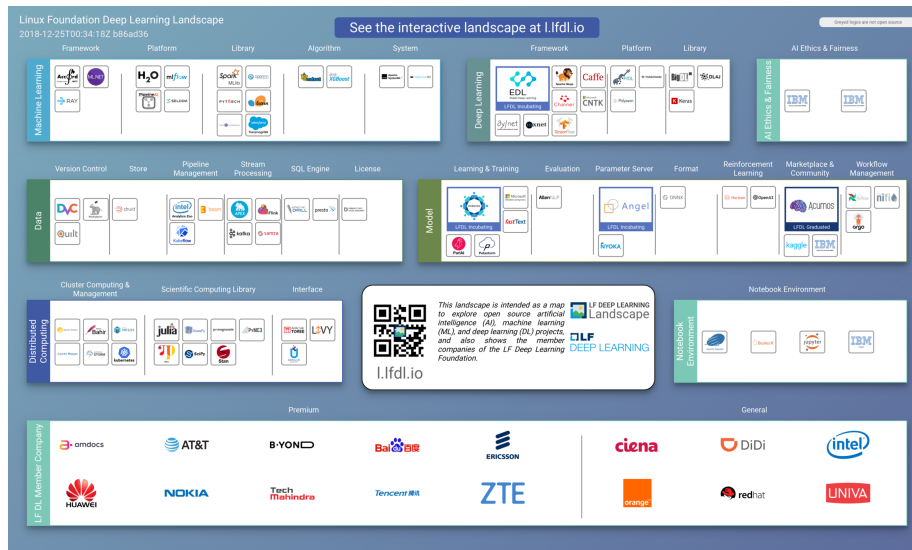
We expect some level of consolidation to happen around frameworks and libraries due to the high level of fragmentation that currently exists. The outcome of this consolidation is likely to be win-win: successful projects will grab their share of additional contributors, and contributors from business as well as academia from older projects have gained the knowledge, experience and skills necessary to meet the demand for talent in this space on new projects.

The LF Deep Learning Foundation

I am excited about what is ahead in the next year and beyond. The LF Deep Learning Foundation anticipates welcoming several new members and open source AI, ML and DL projects in 2019, and continues to focus on supporting and accelerating open source innovation in AI/ML/DL and supporting technical projects within this important space.

Join us!

We recently introduced our first version of the interactive landscape intended as a map to explore open source AI, ML and DL projects. This tool allows viewers to filter, obtain detailed information on a specific project or technology, and easily share via URLs. It is intended to help developers, end users and others navigate the AI, ML and DL open source landscape.



All data is available in a GitHub. We encourage you to spend some time with this tool and contribute to it by submitting a pull request on GitHub.

Contact Information

For any questions in relation to LF Deep Learning Foundation, please send your emails to info@lfdl.io.

